The Geometry of Differential Equations

How equations describe families of curves

A Little Theory

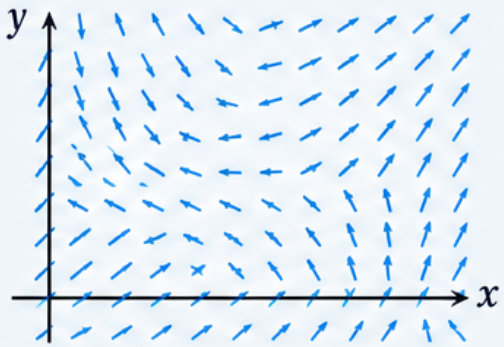
A First-Order ODE

$$\frac{dy}{dx} = f(x, y)$$

assigns a slope at every point in the plane.

$\Longrightarrow$

Direction Field



Solution Curves

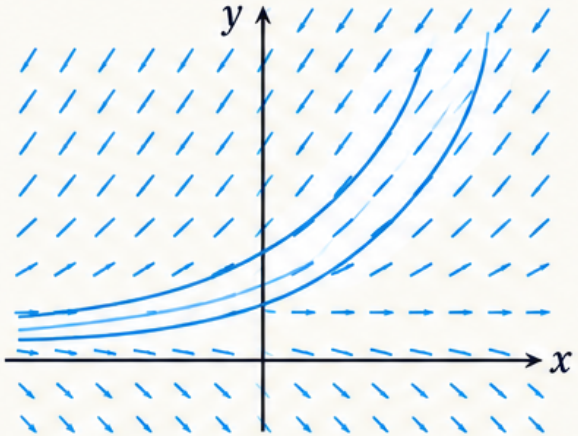
are curves tangent to this field.

Example 1: Exponential Growth

$$\frac{dy}{dx} = y$$

Solutions:

$y = Ce^x$

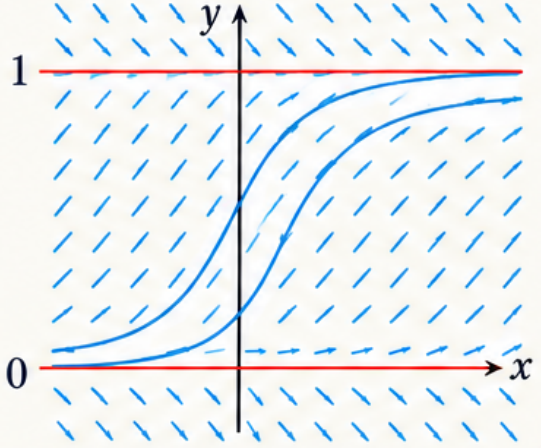


Example 2: Logistic Equation

$$\frac{dy}{dx} = y(1 - y)$$

Equilibrium solutions:

$y = 0, y = 1$



An equation tells us how a curve must move.